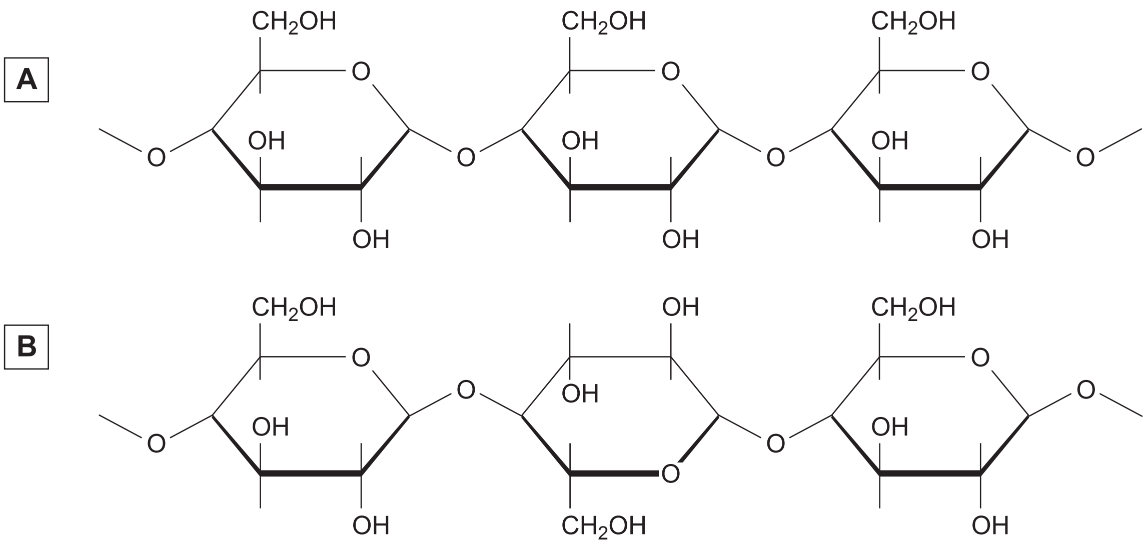
Examiner only

4. (a) Different structural isomers of glucose are found in polysaccharides with very different properties. Parts of two polysaccharides (**A** and **B**) which are found in plants are shown in **Image 4.1**.

Image 4.1



(i) State what is meant by the term structural isomer. [1]

....

....

(ii) Complete **Table 4.2** to describe the structure and function of the polysaccharides **A** and **B**, found in plants. [6]

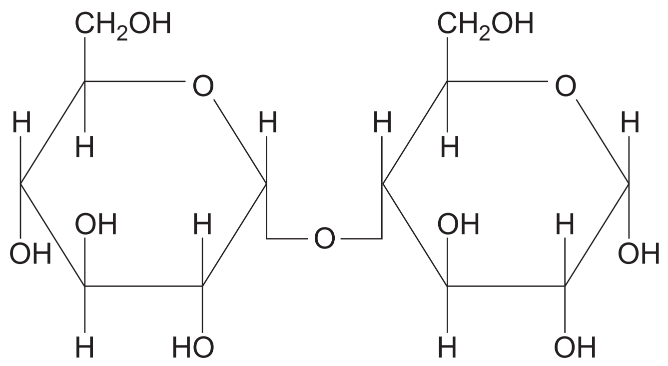
Table 4.2

	Polysaccharide **A**	Polysaccharide **B**
Name of polysaccharide	....	....
Isomer of glucose present	....	....
Description of structure	....	....
Function in a plant cell	....	....

Examiner only

(iii) Polysaccharide **A** is hydrolysed by an enzyme which results in production of the disaccharide shown in **Image 4.3**.

Image 4.3



State the meaning of the term hydrolysis **and** name the disaccharide shown in **Image 4.3**.

[2]

....

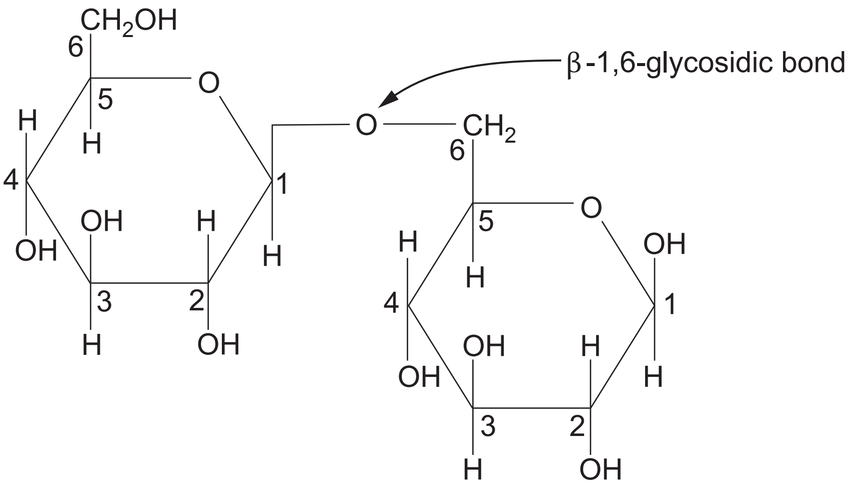
....

....

Examiner only

(b) Gentiobiose, shown in **Image 4.4**, is a disaccharide which consists of two glucose monomers joined by a β-1,6-glycosidic bond.

Image 4.4



Identify **one** structural similarity and **two** structural differences in the disaccharides shown in **Images 4.3** and **4.4**. [3]

Structural similarity ....

....

Structural difference 1 ....

....

Structural difference 2 ....

....

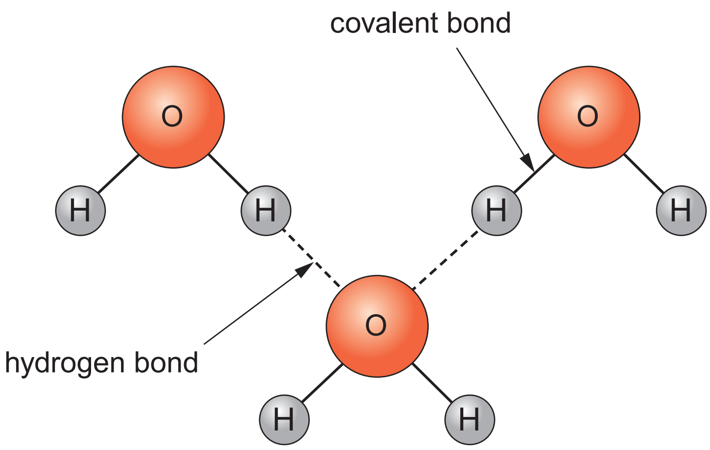
2400U101
15

Examiner only

(c) Monosaccharides and disaccharides are polar molecules, they are soluble in water.

Image 4.5 shows some water molecules and their bonds.

Image 4.5



(i) **Annotate Image 4.5**, to show the polarity of a water molecule. [1]

(ii) With reference to **Images 4.3, 4.4** and **4.5**, explain why monosaccharides and disaccharides are soluble in water. [1]

....

....

....

(iii) State why it is important to plants that sugars are soluble. [2]

....

....

....

....

16